

Chapter 1

Homomorphisms and Isomorphisms

Consider the two groups $\mathbb{Z}_6$ and $\mathbb{Z}_2 \times \mathbb{Z}_3$ (under addition). $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}.$$

These groups have the same number of elements, but as it turns out they have much more in common. Pair the elements as follows:

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (0,0) & (1,1) & (0,2) & (1,0) & (0,1) & (1,2) \end{array}$$

Table 1.1: The elements from $\mathbb{Z}_6$ are mapped to $\mathbb{Z}_2 \times \mathbb{Z}_3$.

This map identifies the zero and identities of each group, $0 \rightarrow (0, 0)$ and $1 \rightarrow (1, 1)$ respectively. Note that in $\mathbb{Z}_6$, $3 + 4 = 1 \pmod 6$ and in $\mathbb{Z}_2 \times \mathbb{Z}_3$, $(1, 0) + (0, 1) = (1, 1)$. Our map identifies $1 \rightarrow (1, 1)$. Our map can be formulated with $(3x + 4y) \pmod 6 \rightarrow (x, y)$ is a function from $f: \mathbb{Z}_6 \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$.

Though these groups appear quite different, this function shows that their behavior is identical. We will call this function an *isomorphism* and two such groups *isomorphic*.

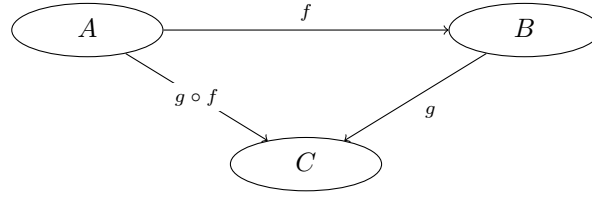
1.1 Functions

We covered functions in MTH 335, but that was a while ago.

Definition 1. A *function* (or *mapping*) from sets A to B , $f: A \rightarrow B$, is a rule that assigns each element in A to exactly one element in B .

- A is called the *domain* of f .
 - B is called the *range* of f .¹
 - If $f(a) = b$ for some $a \in A$, then b is called the *image of a under f*.
 - $\{b: b \in B \text{ and } f(a) = b \text{ for some } a \in A\}$ is called the *image of A under f*.
- Functions are well defined. Meaning that an input is assigned to one and only output.

¹Also called the *co-domain*.

Figure 1.1: Composition of functions f and g .

Example 1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = x^2 + 1$ is a function. it is clear from the equation that f is well defined.

Example 2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = \pm\sqrt{x}$ is not a function. Since $f(1) = +1$ and -1 .

Example 3. Consider $f: \mathbb{R} \rightarrow \mathbb{Z}$ given by the *floor function* $f(x) = \lfloor x \rfloor$ where $\lfloor x \rfloor$ is the largest integer less than or equal to x , e.g., $\lfloor \pi \rfloor = 3$. Note that the floor function is also a function from $\mathbb{R}$ to $\mathbb{R}$, $f: \mathbb{R} \rightarrow \mathbb{Z}$.

Example 4. Let $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x, y) \rightarrow xy$. Then f is a function.

Example 5. Let $A = \{a, b, c, \dots, z\}$ and $f: A \rightarrow \mathbb{Z}$ be function defined by assigning the n^{th} letter to n , e.g., $f(3) = c$.

Definition 2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$. The *composition* of $g \circ f$ is the mapping from A to C defined as follows: $g \circ f(a) = g(f(a))$ for any $a \in A$.

Example 6. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = 2x + 1$ and $g(x) = x^2$. Then $f \circ g(x) = f(g(x)) = 2(x^2) + 1 = 2x^2 + 1$.

Definition 3. A function, $f: A \rightarrow B$, is called *one-to-one*² if $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

Example 7. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ where $f(x) = x^3$ is one-to-one.

Proof. Suppose that $f(x_1) = f(x_2)$ for some $x_1, x_2 \in \mathbb{Z}$. So $(x_1)^3 = (x_2)^3 \Rightarrow ((x_1)^3)^{1/3} = ((x_2)^3)^{1/3} \Rightarrow x_1 = x_2$ □

Exercise 1. Show that any linear function $f: \mathbb{R} \rightarrow \mathbb{R}$ with a non-zero defined slope is one-to-one.

Example 8. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = |x|$ is not one-to-one.

$|-1| = 1 = |1|$ but $-1 \neq 1$ so f is not one-to-one.

Definition 4. A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.

Example 9. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is onto.

Proof. If $b \in \mathbb{R}$ then $\sqrt{b} \in \mathbb{R}$, and $f(\sqrt{b}) = (\sqrt{b})^2 = b$. □

Example 10. $f: \mathbb{Z} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is not onto.

Proof. $2 \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Z}$. □

Exercise 2. Prove that the floor function (see example 3) is onto but not one-to-one.

²Also called an *injective function*

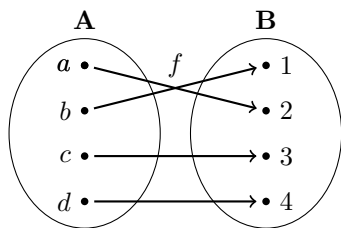


Figure 1.2:

$f: A \rightarrow B$ is one-to-one and onto.

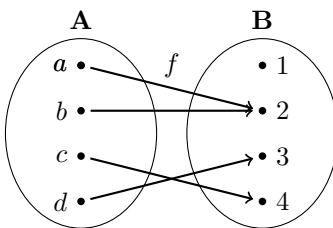


Figure 1.3:

$f: A \rightarrow B$ is not one-to-one and is not onto.

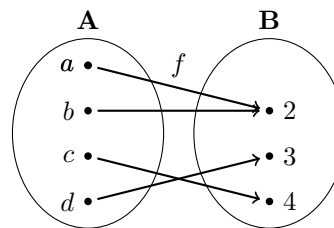


Figure 1.4:

$f: A \rightarrow B$ is onto but not one-to-one.

1.2 Homomorphism

Definition 5. A (ring) *homomorphism*, f , from a ring R to a ring S is a function $f: R \rightarrow S$ that preserves the two ring operations:

$$f(a + b) = f(a) + f(b) \text{ and } f(ab) = f(a)f(b)$$

for $a, b \in R$.

Example 11. Consider the function $f: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ given by $f(x) = x^2$. f is a homomorphism.

$$f(a + b) = (a + b)^2 = a^2 + 2ab + b^2 = a^2 + b^2 \text{ since } 2ab = 0 \pmod{2}, \text{ and } f(ab) = (ab)^2 = a^2b^2 = f(a)f(b).$$

Exercise 3. Prove that $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $k \rightarrow k \pmod{n}$ is a ring homomorphism.

Example 12. The map $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x$ is not a homomorphism.

While the operation addition is preserved, $f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$, the multiplication operation is not, $f(ab) = 2ab \neq f(a)f(b) = 4ab$.

Exercise 4. Prove that the floor function (see example 3) mapping $\mathbb{R}$ to $\mathbb{Z}$ is not a homomorphism.

Exercise 5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2^x$. Show that f is not a homomorphism under the usual operations, but that is a homomorphism under addition and multiplication defined as follows: $2^x \oplus 2^y = 2^{x+y}$ and $2^x \otimes 2^y = 2^{xy}$.

Exercise 6. Prove that every ring homomorphism $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ has the form $f(x) = ax$ where $a^2 = a$. Hint: look at the identity; call $f(1) = a$.

If $f: R \rightarrow S$ is a homomorphism and $g: S \rightarrow T$ is a homomorphism, is $g(f): R \rightarrow T$ a homomorphism? Prove your answer.

Exercise 7. Is the map $f: \mathbb{Z}_8 \rightarrow \mathbb{Z}_8$ given by $f(x) = 2x$ a ring homomorphism? Prove your answer.

Exercise 8. Is the mapping $f: \mathbb{Z}_5 \rightarrow \mathbb{Z}_{30}$ given by $f(x) = 6x$ a ring homomorphism? Prove your answer.

Theorem 1. Properties of Ring Homomorphisms. Let f be a ring homomorphism from rings R to S , and let A be a subring of R .

1. $f(0_R) = 0_S$.
2. $f(-a) = -f(a)$ for every $a \in R$.

3. $f(a - b) = f(a) - f(b)$ for every $a, b \in R$.
4. For any $a \in R$ and any positive integer n , $f(nr) = nf(r)$ and $f(r^n) = f(r)^n$.
5. If R is commutative then $f(R)$ is commutative.
6. $f(A) = \{f(a) : a \in A\}$ is a subring of S .
7. If R has unity 1_R , $S \neq \{0\}$, and f is onto
 - (a) then $f(1_R)$ is the unity of S .
 - (b) if $aa^{-1} = 1_R$ for any $a, a^{-1} \in R$, then $f(a)f(a^{-1}) = 1_S$ and $f(a)^{-1} = f(a^{-1})$.

Exercise 9. ★ Prove properties 1, 2, and 3 of theorem 1.

Exercise 10. ★ Prove properties 4 and 5 of theorem 1.

Exercise 11. ★ Prove property 6 of theorem 1.

Exercise 12. ★ Prove properties 7.a and 7.b of theorem 1.

1.3 Isomorphisms

Definition 6. A *ring isomorphism* from a ring R to a ring S is a function that is one-to-one and onto.

Definition 7. If there exists a ring isomorphism $f: R \rightarrow S$ then R and S are said to be *isomorphic*.

Showing that two rings are isomorphic usually requires finding an isomorphism between the, and this can be tricky at times. However looking at the properties from theorem 1 can serve as a guide.

Example 13. The function shown in table 1.1 is an isomorphism. The function $f: (x, y) \rightarrow (3x + 4y) \bmod 6$ is an isomorphism.

Example 14. The function $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ from exercise 3 is not an isomorphism.

f is a homomorphism (exercise 3) and it is onto since $f(a) = a$ for any $a \in \mathbb{Z}_n$. However it is not one-to-one; consider $f(nk) = 0 \bmod n$ for any $k \in \mathbb{Z}$.

Exercise 13. Show that the rings $\mathbb{Q}$ and $\mathbb{R}$ are not isomorphic to $\mathbb{Z}$.

Exercise 14. Let $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$. Prove that $f: \mathbb{Q}[\sqrt{2}] \rightarrow \mathbb{Q}[\sqrt{2}]$ defined by $f(a + b\sqrt{2}) = a - b\sqrt{2}$ is an isomorphism.

Example 15. Prove that the $n\mathbb{Z}$ is ring isomorphic to $m\mathbb{Z}$ if and only if $m = n$. Recall that $n\mathbb{Z}$ is the ring of all n integer multiples.

Exercise 15. Show that any ring isomorphism, $f: \mathbb{Z} \rightarrow \mathbb{Z}$, must be the identity map.

Recall that a function $f: A \rightarrow B$ can be one-to-one only if $|A| \leq |B|$. That is, A having fewer elements than B is necessary condition for f to be one-to-one. Also, $|B| \leq |A|$ is a necessary condition for f to be onto. Clearly, $|A| = |B|$ is a necessary condition for f to be an isomorphism.